

Tomas Ulises Traini

Software Engineer — Clinical Research Systems

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SUMMARY

Full-stack software engineer, ~5 years, .NET/C# and React, now Python/FastAPI. Sole engineer on ClinigmaScriber at CLINIGMA ApS (Denmark): a clinical-research transcript system that takes interview audio through transcription, translation and qualitative coding on a git-backed, append-only versioning model built around GCP/GxP and 21 CFR Part 11 expectations. I chose the architecture, the stack and the coding standards. Before that: React frontend and SAP-backed microservices at John Deere, .NET modernization and application security at PwC. Argentine and Spanish (EU) citizen. English C2, German B1 (B2 in progress).

SKILLS

Languages: Python, TypeScript, C#, JavaScript, SQL

Backend: FastAPI, SQLAlchemy (async), Pydantic, Alembic, .NET / .NET Core / .NET Framework, NestJS, Node.js, REST, GraphQL, WebSockets

Frontend: React, TypeScript, Zustand, TanStack Query, Redux, Vite, Tailwind, Lexical, ECharts, i18next, Angular

Data: PostgreSQL, SQL Server, DynamoDB, JSONB, full-text search (tsvector/trigram), SAP integration

Cloud & infrastructure: Azure (App Service, Blob, Cosmos DB for PostgreSQL, OpenAI, Translator, Speech), AWS (SQS, load balancing), Terraform, Docker, CI/CD, Azure DevOps, GitHub Actions

Architecture & quality: hexagonal (ports & adapters), microservices, pytest, vitest, Cypress, Playwright, WCAG 2.1, Checkmarx

Clinical / research domain: ICH-GCP, GxP, 21 CFR Part 11 concepts, audit trails, e-signature workflows, qualitative coding, codebooks, QDPX (REFI-QDA qualitative project exchange), ASR, speaker diarization

AI: agentic development workflows, multi-agent orchestration, Azure OpenAI, Claude CLI, GitHub Copilot, Cursor

EXPERIENCE

Technical Officer — Software Engineer (Consultant, self-employed)

CLINIGMA ApS · Denmark, Remote · Oct 2025 – Present

CLINIGMA runs in-trial patient interviews and produces qualitative patient-experience data for pharma research and regulatory submissions, under ICH-GCP and ISO 9001.

- **Sole engineer on ClinigmaScriber** since Apr 2026 — architecture, technology choices and coding standards were mine, refined in review with the CTO. Python 3.11 / FastAPI backend in a strict hexagonal layout (ports & adapters, enforced by import-linter) with a React + TypeScript frontend (Vite, Zustand for client state, TanStack Query for server state, Lexical for the transcript editor), on PostgreSQL 16. UI localized in Danish, German, English and Spanish.
- **Built the three-stage clinical workflow** — transcription → translation → coding, each gated by a trial manager's sign-off — on a sentence-addressable, append-only versioning model backed by embedded git (one bare repo per interview via dulwich). Designed around GCP/GxP and 21 CFR Part 11 expectations; a Part 11-*shaped* substrate, not a validated or certified system.
- **Audio pipeline.** Azure Fast Transcription in production for ASR with word-level timestamps, speaker diarization and language auto-detect. Also built a local CLI with pluggable backends (Whisper, Qwen3, pyannote, NeMo Sortformer), forced alignment, WER and timing-drift scoring, and a bake-off harness for comparing backends on the same audio.
- **Translation lane** freezes a byte-exact copy of the locked source transcript and seeds a machine-translation draft. On a source re-lock, only changed sentences re-translate — every human-reviewed sentence carries forward byte-identical.
- **Coding stage is text-frozen:** coders write annotations only, anchored to sentence spans, against a project codebook (tree structure, ownership, labels immutable after first use). Codebook exchange ships as a JSON envelope today, with QDPX (REFI-QDA qualitative project exchange) / .qdc interop as the committed next step for ATLAS.ti, NVivo and MAXQDA.
- **Provenance and gated AI.** Every published change is a commit plus an append-only log row plus a reviewer identity; every model call is audit-logged with model, version, prompt hash and input/output hashes. AI features ship closed behind per-deployment flags and produce proposals a human disposes of, never a source of truth.
- **Held the engineering discipline** on a codebase covered by ~4.5k backend and ~2.8k frontend tests plus Cypress end-to-end twins, with ~20 mechanical CI gates I defined (500-line file cap, layer-import rules, cross-page import rules, a React Query key ledger, locale parity) and a binding bug-precedent ledger. Deployed to Azure App Service via Azure DevOps.
- **Owned and then killed** a schema-driven compliance-document system: reusable schema units and forms so structures could be reused across many documents in a project. Stopped it when testing showed the multi-use vs single-use decisions were too complex for users whose baseline tool is Word.
- **Built experimental agentic development workflows** with my manager: a board-discussion system (brief → specialist agents discuss → orchestrator synthesizes the thread) and a ticket flow (GitHub ticket → Teams refinement with humans → idea refiner → architect → one implementation agent per stack → QA with a retry loop → PR creation), plus a separate flow for code review. These experiments informed a large legacy jQuery-to-React rewrite of the CLINIGMA portal led by my manager.
- **Status:** the engine covers the full lifecycle; first internal pilot on mock interviews begins Sep 2026. Portal integration is designed, not yet delivered.

Senior Software Developer — Sistran Consultores

Remote · Aug 2025 – Feb 2026

- Built React and .NET features for insurance policy workflows inside an established enterprise codebase and delivery process.

Ssr. Software Engineer — John Deere

Remote · Sep 2023 – Jun 2025

- Primary React/TypeScript engineer on web applications for tractor parts management, in a Node.js microservices architecture with Terraform and CI/CD; worked directly with product to turn user stories into shipped features.
- Part of a major frontend refactor: replaced prop drilling with reducer-based state, standardized TypeScript conventions (removing any and over-generic types), and fixed inconsistent naming across the codebase.
- Implemented GraphQL queries and pagination for sorting large parts catalogues, where a single SAP query could return ~100,000 rows and naive rendering broke the UI; performance was the binding constraint.
- Owned features end to end when they crossed into SAP: connected the parts service to SAP, coordinated the SAP counterpart so their events and endpoints landed in parallel, extended the events service (AWS SQS, load balancing) and wired live part-state updates through to the frontend via WebSockets — handling idempotency, race conditions and states that diverged between SAP and our system. Documented each flow as use cases and handed it to QA.

ReactJS Developer — Podium Pick'em Challenge (Contractor)

Remote · Apr 2024 – Jun 2024 (*freelance project, concurrent with John Deere*)

- Built a responsive React frontend with reusable components for a sports prediction and betting platform tied to the 2024 Olympics, under a tight deadline.

Ssr. .NET Developer — PwC

Remote · Aug 2022 – Sep 2023

- Co-led a 3-person team modernizing legacy .NET Framework systems for US clients; the team was recognized by visiting US leadership for a substantial year-on-year reduction in open vulnerabilities.
- Acted as the reference person for application security: resolved 30+ Checkmarx findings and engineered staged .NET and NuGet upgrade paths for codebases with known-vulnerable packages, resolving version conflicts without rewriting whole systems; escalated one blocking package issue to Microsoft's internal team.
- Delivered features, maintenance and WCAG 2.1 accessibility work for PwC Policy on Demand (policyondemand.pwc.com), a policy video platform still in production.
- Implemented Excel-generation reporting for tax-calculation projects in a compliance-heavy area.

.NET Developer — Ingenea S.R.L.

Hybrid, Rosario · Dec 2021 – Aug 2022

- Built and maintained .NET Core APIs and SQL Server stored procedures for clients across Argentina, Brazil and Mexico, including Ternium S.A., structuring data access through repository and service layers per client.

EDUCATION

Universidad Abierta Interamericana

Bachelor's Degree in Technology Information Management (Licenciatura en Gestión de Tecnología Informática) · Apr 2025 – Expected Nov 2027 (in progress)

IT governance, project management, enterprise systems.

Instituto Zona Oeste

Technical Degree in Software Development (3-year technical program) · Mar 2019 – Nov 2022 · GPA 7.33

Software engineering fundamentals, .NET Core, C#, Angular, SQL Server.

LANGUAGES

- Spanish — Native
- English — C2
- German — B1 (B2 in progress)